HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 21 Name: _____ Marks: / 10

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If c and d are constants such that $(x+2)(x+c)+12 = x(x+d)+6c(x+1)$, then $d =$

   A. -13.

   B. -3.

   C. 3.

   D. 17.

(DSE-2023 PII Q05)

Answers written in the margins will not be marked.

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2. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

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3. The equations of the circles C_1 and C_2 are $x^2+y^2+7x-4y+15=0$ and $2x^2+2y^2-2x-16y-17=0$ respectively. Let G_1 and G_2 be the centres of C_1 and C_2 respectively. Denote the origin by O. Which of the following are true?

I. $\triangle OG_1G_2$ is an equilateral triangle.

II. The line segment OG_1 lies inside C_2.

III. C_1 and C_2 intersect at two distinct points.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q27)

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4. The L.C.M. of a^4b^2c, a^3b^4c and $a^2b^5c^2$ is

A. a^2b^2c.

B. $a^2b^2c^2$.

C. a^4b^5c.

D. $a^4b^5c^2$.

(DSE-2023 PII Q32)

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5. The equations of the straight lines L_1 and L_2 are $3x-4y+k=0$ and $4x+3y-k=0$ respectively, where k is a positive constant. It is given that L_1 cuts the x-axis at the point P. Denote the point of intersection of L_1 and L_2 by Q. If R is a point lying on L_2 such that the in-centre of $\triangle PQR$ lies on the x-axis, then the x-coordinate of R is

A. $-7k$.

B. $-k$.

C. k.

D. $7k$.

(DSE-2023 PII Q41)

I can finish _____ questions

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